

Adult body weight modifies geroprotector efficacy in genetically heterogeneous mice

Analysis

Short title: Adult weight and geroprotector efficacy

Keywords: body weight; adiposity; lifespan; geroprotector; Interventions Testing Program; UM-HET3; Diversity Outbred; effect modification

Abstract

Mouse lifespan studies express treatment efficacy relative to ad-libitum-fed controls, whose metabolic state may itself constrain survival. Whether efficacy depends on that state has not been tested within arms. We harmonized individual body weights and survival for 26,717 mice across 89 randomized comparisons of the NIA Interventions Testing Program (ITP) and fitted treatment-by-weight interactions. In males, each standard deviation of attained adult weight made treatment approximately 5% more favourable (HR 0.95, 95% CI 0.91–0.99; $P = 0.012$), an effect concentrated among weight-lowering interventions (HR 0.90, 0.85–0.94). Median lifespan gain rose from 6.6% in the lightest pre-treatment quarter to 20.4% in the heaviest, mainly because control survival declined more steeply with weight than treated survival. The same ordering recurred for 12 of 13 compounds, with the lightest animals retaining only about one-quarter of the gain observed in the heaviest. The modifier was weight attained during adulthood rather than early-life size, while the attenuated female association was specific to the ITP setting. Large effects against heavy, short-lived controls are therefore best interpreted as partial rescue conditional on comparator metabolic state, and may overstate benefit in metabolically healthier populations.

Main

Mouse lifespan studies usually express efficacy as a percentage difference from an ad-libitum-fed control group, but the biological meaning of that contrast depends on the condition of the comparator. Continuous access to food under sedentary housing can produce excess adiposity, insulin resistance and shortened survival relative to a metabolically healthy state. An intervention can therefore generate a large relative lifespan gain either by raising the survival ceiling or by rescuing pathology concentrated in a vulnerable control population.¹

This distinction matters for translation. Preventive geroprotectors may be used across a broad range of adults, but effects driven largely by reversal of weight-associated disease in heavy mice may translate less readily to healthy-weight individuals. Associations between weight loss and lifespan gain across treatment arms raise this possibility,² and smaller treatment effects have been reported in longer-lived ITP control cohorts,² but neither can show whether the same randomized intervention works differently in lighter and heavier animals. That question requires an individual-level treatment-by-weight analysis.

The NIA Interventions Testing Program (ITP) provides a comprehensive resource for this test. UM-HET3 mice are genetically heterogeneous, interventions are randomized, and cohorts are replicated across three sites using standardized husbandry and survival ascertainment.³ GeneNetwork adds harmonized longitudinal, maternal and genetic traits. Together, these resources distinguish early-life size from attained adult weight and allow us to test within randomized arms whether treatment effects vary with body weight.

We therefore asked whether adult body weight modifies survival benefit across interventions; whether lifespan benefit increases across the pre-treatment weight distribution; whether the pattern reflects early-life size or attained adult weight; and whether sex, genetic architecture or husbandry help explain heterogeneity.

Results

Adult weight modifies male treatment efficacy

The harmonized resource included 26,717 unique mice across 89 randomized comparisons and 16 cohorts (14,301 males and 12,416 females). To compare animals within each experiment, the earliest usable adult

weight was standardized within treatment group, site and sex. This removed average weight differences caused by the intervention and tested whether relatively heavier animals within the same randomized group received a different survival benefit.

Across 89 male arms, each additional standard deviation of landmark weight made the relative treatment effect approximately 5% more favourable (HR 0.95, 95% CI 0.91–0.99; $P = 0.012$; 14,301 mice). Females showed no corresponding moderation across 85 arms (HR 1.02, 0.98–1.06; $P = 0.29$; 12,416 mice). In males, the association was concentrated among weight-lowering interventions: each additional standard deviation of weight made treatment approximately 10% more favourable in 39 weight-lowering arms (HR 0.90, 0.85–0.94; $P = 3.6 \times 10^{-5}$), compared with no clear moderation in 50 non-lowering arms (HR 0.99, 0.95–1.04; $P = 0.81$). Fitting both classes in one model showed a stronger treatment-by-weight association for weight-lowering interventions (ratio of interaction HRs 0.90, 95% CI 0.85–0.95; $P = 2.9 \times 10^{-4}$), whereas neither female class showed clear moderation. Restricting weight to measurements taken before dosing gave the same pattern: HR 0.91 (0.85–0.96; $P = 0.0011$; 27 arms) for weight-lowering interventions and 1.02 (0.96–1.08; $P = 0.49$; 29 arms) for non-lowering interventions, with a class contrast of 0.89 ($P = 0.0014$). Arm-level estimates and the set of moderation models are shown in Fig. 1a,b and Table 1.

Treated animals converge while heavy controls account for much of the headline gain

Median lifespan showed the same interaction directly. Across 27 male weight-lowering arms with weight measured before dosing, quartiles were formed separately within cohort, site, intervention and treatment group, then standardized so that the same comparison cells contributed equally to every quarter. Median lifespan gain rose from 59 days in the lightest quarter (892 to 951 days; +6.6%; mean control weight 34.8 g) to 137 days in the heaviest (673 to 810 days; +20.4%; 47.1 g), a 3.1-fold difference in proportional gain (Fig. 2a; Table 2). The second and third quarters gained 10.1% and 12.6%. A Cox model confirmed a monotonic gradient (HR 0.92 per heavier quarter, 95% CI 0.87–0.97; $P = 0.0011$). From the lightest to the heaviest quarter, control median lifespan fell by 219 days, compared with 141 days in treated animals—a 78-day attenuation of the weight-associated decline. Conventional site-pooled medians gave the same ordering (5.2%, 12.7%, 14.5% and 19.5%), a 3.8-fold lightest-to-heaviest contrast (Fig. 2b).

The gradient was specific to interventions that lowered adult weight: corresponding gains across the four quarters were only 4.2%, 1.1%, 4.4% and 4.7% in 29 non-lowering male arms. The relation also saturated rather than increasing without limit. Narrowing the heaviest stratum from a third to a quarter to a fifth raised mean control weight from 46.6 to 47.9 g, while gain remained 20.6%, 19.5% and 19.3%. This plateau is consistent with treated survival approaching a narrower upper range while untreated survival continues to decline.

Even the lowest-weight quarter had a mean control weight of 34.8 g, approximately 15% above the 30.3 g mean across 27 male laboratory strains measured at a similar age.⁴ Thus, the lowest-weight quarter represents a relative weight stratum rather than a metabolically lean or low-adiposity reference (Fig. 3a).

The weight gradient recurs across compounds

To test whether the pooled gradient was confined to a small number of interventions, we collapsed the 24 single-agent arms onto their 13 active compounds. Across the interventions, the lowest-weight animals retained only one-quarter of the lifespan gain observed in the heaviest animals (5.2% versus 20.9%). The same ordering was observed for 12 of the 13 compounds, including acarbose (7.3% versus 28.4%), 17 α -estradiol (5.9% versus 22.9%), canagliflozin (6.1% versus 23.2%) and rapamycin (6.7% versus 11.6%).^{5–9} The recurrence across pharmacologically distinct compounds indicates that the pooled association is not carried by a single intervention or dosing schedule. The corresponding common treatment-by-weight interaction was HR 0.91 (95% CI 0.85–0.97; $P = 0.0022$) (Table 3).

Adult weight predicts male mortality across untreated cohorts

In untreated controls, greater six-month weight predicted substantially higher subsequent mortality in males (HR 1.39 per SD, 1.34–1.44; $P = 2.3 \times 10^{-79}$) and a smaller increase in females (HR 1.08, 1.04–1.11; $P = 1.4 \times 10^{-5}$; Fig. 4a; Extended Data Table 1). The direction was nearly universal across cohort $\times$ intervention strata: six-month weight was inversely associated with lifespan in 104 of 105 male strata (median Spearman $\rho = -0.30$), and 83 of 101 female strata (median $\rho = -0.07$). The difference in magnitude persisted at twelve

months (male HR 1.36; female HR 1.09; Fig. 4a) and weakened by eighteen months. By 24 months the male association had reversed, consistent with selective survival and weight loss before death. The six-month gradient was unchanged after adjustment for maternal age and parity, both estimates shifting by less than 0.01 (Extended Data Table 2).

The signal emerged with acquired adult weight rather than early-life size

Body weight at puberty (approximately 1.4 months of age) was available for 3,568 animals across three cohorts, including 2,839 with a six-month measurement. It did not modify treatment efficacy in either sex (males: four arms, 730 mice, HR 1.00, 0.85–1.17; $P = 0.97$; females: two arms, 553 mice, HR 1.03, 0.86–1.22; $P = 0.77$; Fig. 4b). Among the 539 male controls with both measurements, 1.4-month weight was not independently associated with mortality after six-month weight was included in the same model (HR 0.97 per s.d., 0.87–1.07; $P = 0.52$), whereas six-month weight remained strongly associated with mortality (HR 1.46, 1.31–1.62; $P = 2.7 \times 10^{-12}$; Fig. 4c).

Repeated-weight models identify acquired adult weight as the relevant risk state

We next asked whether weight gained during adulthood carried additional risk beyond weight already attained by six months. After adjustment for six-month weight, greater weight gain from six to twelve months predicted higher mortality in males (HR 1.09, 1.06–1.12; $P = 2.4 \times 10^{-10}$; 11,300 mice across 207 cohort $\times$ site $\times$ arm cells), with a smaller association in females (HR 1.03, 1.01–1.05; $P = 9.0 \times 10^{-4}$; 10,521 mice across 201 cells). The male association remained when the analysis was restricted to controls and interventions beginning at or after twelve months (HR 1.09, 1.05–1.13; $P = 2.4 \times 10^{-6}$; 5,393 mice across 81 cells). Together with the 1.4-month analysis, these results indicate that the relevant phenotype is attained adult weight rather than constitutional body size.

Heavier comparator groups yield larger treatment-effect estimates

A complementary arm-level analysis grouped 426 cohort $\times$ site $\times$ sex $\times$ arm cells according to the six-month weight of their own control group. In males, pooled hazard reduction halved from 22.3% against the heaviest comparator tertile to 11.0% against the lightest, and mean median-lifespan gain decreased from 9.3% to 4.1% (Table 4). The pattern remained when comparisons were made within cohorts ($P = 0.0081$) and was attenuated in a pooled sensitivity analysis with variance clustered on the 39 shared cohort $\times$ site control groups ($P = 0.078$). Female estimates showed the same descriptive ordering, with hazard reduction decreasing from 15.1% to 4.3% and median-lifespan gain from 4.1% to 1.6%, but neither analysis established a within-population gradient ($P = 0.63$ and $P = 0.37$).

Changes in body-fat percentage do not track lifespan extension

We then asked whether changes in body-fat percentage could explain the variation in lifespan benefit. Across 36 lifespan-extending arms from 14 studies, the treatment–control difference in body-fat percentage was not associated with lifespan gain (Pearson $r = -0.10$, 95% CI -0.41 to 0.24 ; $P = 0.58$; Spearman $\rho = 0.08$; $P = 0.66$). The same conclusion held when results were aggregated by study, analysed with study-clustered regression or assessed by leave-one-study-out analysis.^{10–15} Individual interventions also showed discordant fat and lifespan responses (Extended Data Table 3). Changes in body-fat percentage therefore did not track lifespan gain across these arms.^{16,17}

The attenuated female association is concentrated in the ITP setting

Within ITP arms in which treatment both lowered body weight and extended lifespan, each 1% reduction in weight corresponded to a fitted lifespan gain of 1.03% in males (95% CI 0.75–1.32; 31 arms) but only 0.46% in females (0.17–0.75; 25 arms). The female slope was therefore 0.57 percentage points lower than the male slope (95% CI 0.16 to 0.98 lower; $P = 0.0063$). Outside ITP, however, the corresponding pharmacological slopes were similar in males and females: 0.79% (0.40–1.18; 29 arms) and 0.93% (0.55–1.31; 22 arms; $P = 0.54$). Dietary-restriction slopes were likewise similar between sexes (0.81% in males and 0.79% in females; $P = 0.90$). The pharmacological sex-by-source interaction was significant ($P = 0.021$) and remained so after restricting the analysis to arms with at least 5% weight loss ($P = 0.018$). The attenuated female association

was therefore concentrated within ITP rather than observed consistently across mouse intervention studies (Extended Data Table 4).

Population architecture may contribute to the ITP-specific pattern

Some features of the ITP setting could bear on the weaker female association. UM-HET3 mice are genetically heterogeneous and descend from four inbred founders, while other cohorts in the sampled studies such as Diversity Outbred mice derive from eight founders, including three wild-derived strains and strains susceptible to obesity or autoimmune diabetes.^{18–21} This broader genetic architecture may encompass a wider range of metabolic susceptibility, allowing adult weight to index accumulated host vulnerability differently across populations (Extended Data Table 5).

ITP controls also occupied a substantially heavier and more adipose adult state than their founder strains. In an independent Mouse Phenome Database survey, males of the four UM-HET3 founders weighed an average of 29.4 g at approximately 7 months (26.7–32.0 g; Extended Data Table 6), while untreated ITP males averaged 40.4 g at six months—approximately 37.5% heavier despite being measured 1 month earlier. Even the lowest-weight ITP quarter averaged 34.8 g, exceeding every founder strain. Although heterosis may explain part of this difference in body size, it less clearly explains the accompanying adiposity: percentage fat averaged 35.8% in male and 43.4% in female ITP controls, compared with founder means of 25.1% and 20.9%, respectively. Only two of the 40 surveyed strains reached the male ITP value and one reached the female value (Fig. 3b,c).^{4,22,23}

Continuous access to food within sedentary laboratory housing may contribute to this metabolic state.¹ In UM-HET3 mice, controlled food availability lowered male weight from approximately 45–47 g to 25–29 g while increasing mean lifespan from 831 to 1,101 days (+32%) and female lifespan from 836 to 1,169 days (+40%).²⁴ These coordinated differences in weight and survival could indicate that part of the observed longevity benefit may reflect recovery of survival lost under an adverse metabolic state rather than raising lifespan from a metabolically healthy baseline.

Discussion

We find that reported geroprotective efficacy in male mice depends materially on attained adult weight. Treatment-by-weight moderation was present across 89 randomized arms and was concentrated among interventions that lowered weight. Expressed as median lifespan, benefit increased approximately threefold from the lightest to the heaviest pre-treatment quarter. This gradient arose mainly because control survival declined more steeply with weight than treated survival. The same dependence was evident at the compound level, where lifespan extension was approximately four times as large in the heaviest weight quarter as in the lightest, with the same ordering for 12 of 13 compounds. Together, these patterns are consistent with partial rescue of weight-associated risk rather than uniform elevation of a common lifespan ceiling.

This distinction changes the translational interpretation. The 20.4% gain in the heaviest quarter and the 6.6% gain in the lightest quarter reflect different starting conditions, not a single context-free treatment effect. ITP mice are maintained with unrestricted access to food under sedentary laboratory conditions and remain relatively heavy even in the lightest quarter. Its control median lifespan—892 days—was closest to the approximately 900-day reference proposed for distinguishing adequately long-lived from short-lived control groups, whereas controls in the heaviest quarter reached 673 days, 25% below the 900-day reference.² Thus, the lightest quarter may represent the closest available approximation to a lower-risk baseline, although it may not itself be representative of metabolically lean mice or controls under controlled food availability. This interpretation accords with longstanding concerns that continuously fed, sedentary laboratory rodents can develop excess weight, impaired glucose regulation and premature mortality, and that interventions effective in this state may reproduce less strongly in active, normal-weight populations.¹

The associated phenotype appears to reflect weight acquired during adulthood rather than constitutional body size. Early-life weight predicted mortality in isolation but was not independently associated once adult weight was included, whereas six-month weight remained strongly predictive and subsequent adult gain added risk. The relevant treatment modifier is therefore better described as attained adult weight than as body size established early in life.

The sex difference was concentrated in ITP rather than shared across mouse intervention studies. Within ITP, male control mortality rose steeply with adult weight, whereas the female gradient was smaller. Outside ITP,

lifespan gain scaled similarly with weight reduction in both sexes. This pattern is more consistent with an ITP-specific population, intervention mix or husbandry effect than with a broadly attenuated association in females.

Body weight reflects fat mass, lean mass, organ size, hydration, fat distribution and cumulative energy balance,²⁵ while percentage fat can remain unchanged when fat and lean mass decline in parallel. The data point to effect modification by a broader adult physiological state indexed by body weight rather than by adiposity alone. Changes in body-fat percentage were not associated with lifespan gain, leaving adiposity as a possible component of that state rather than an established mediator.

These findings suggest several priorities for future experiments. Studies would benefit from reporting effects across baseline metabolic strata and from testing leading interventions in both metabolically healthy and metabolically challenged comparators. Pair-fed or weight-matched controls could distinguish effects reproduced by altered energy balance from benefits that exceed it. Replication across genetically distinct populations representing the breadth of human metabolic susceptibility would test whether the findings generalize across genetic backgrounds and physiological states. These analyses concern variation within ad-libitum-fed populations and do not determine how far the ad-libitum state itself is displaced from a metabolically healthy baseline. Virtually all sampled lifespan experiments compared treatment against ad-libitum-fed controls, whereas survival effects against metabolically healthy, non-ad-libitum comparators remain largely untested. The mouse data suggest that efficacy may not be fully maintained as baseline metabolic risk falls. Reliance on heavy, short-lived controls may therefore bias expected benefits upward when treatment effects estimated against ad-libitum controls are extrapolated to metabolically healthier populations.^{1,26,27}

In summary, attained adult weight modified male geroprotective efficacy in the ITP, with larger effects in heavier animals and for weight-lowering interventions. Large effects against heavy, short-lived controls are therefore best interpreted as evidence of partial rescue conditional on comparator state until efficacy is also demonstrated against metabolically healthy controls or under controlled-feeding conditions.

Methods

Data sources and harmonization

Individual body weights were combined from Mouse Phenome Database cohorts 2010, 2011, 2013, 2016, 2020 and 2021 and GeneNetwork HET3-ITP cohorts 2004–2007, 2009, 2014, 2015 and 2017–2019, yielding 36,973 unique animals, of which 27,637 had at least one recorded adult weight.^{3,28} Cohort 2012 supplied survival records but no adult-weight measurements and therefore did not enter the weight-based models. The 4,430 records present in both sources had identical weight values. Sex was taken from lifespan files because the 2016 phenotype file mislabelled 787 animals at one site. Identifiers were normalized across source conventions, duplicate 2020 lifespan records were removed, and weights outside 10–80 g were treated as recording errors. The primary treatment-moderation model retained 26,717 unique mice after arm and landmark eligibility. Death and censoring definitions followed the source files; controls reused across intervention comparisons were retained with variance clustered by mouse.

Primary all-arm treatment-moderation model

The primary analysis included every intervention arm with treated and control animals, at least 60 mice in total, at least 40 deaths, survival follow-up after the weight landmark and a usable adult weight. For arms dosed after six months, six-month weight was used; for arms dosed by six months, twelve-month weight was used. Weight was z-standardized within treatment group $\times$ site $\times$ sex. Cox follow-up began at the landmark, baseline hazards were stratified by arm $\times$ site $\times$ sex, treatment and weight main effects were arm-specific, the treatment-by-weight coefficient was shared, and robust variance was clustered by mouse. HR < 1 indicates a more favourable treatment effect at higher within-arm weight. Interventions were classified as weight-lowering when treated animals averaged more than 1 g below their own controls across available post-dosing adult measurements, weighted by treated sample size. This classifier used weight response rather than survival outcome. Class differences were tested in joint Cox models containing treatment-by-weight-by-class terms with mouse-clustered variance. Because the primary landmark could follow treatment initiation in early-start arms, the class analysis was repeated using only modifier weights measured before dosing.

245 Secondary successful-arm landmark models

246 A secondary analysis was restricted to intervention–sex combinations reported to extend lifespan in the
247 corresponding ITP publication. Models were fitted at 6, 12, 18 and 24 months, with emphasis on twelve
248 months as the best compromise between established adult weight and limited mortality selection. A 90-day
249 lag excluded deaths shortly after each landmark. Leave-one-cohort-out analyses assessed sensitivity to
250 individual cohorts.

251 Comparator-state analyses

252 For the absolute-survival analysis, 27 male weight-lowering arms with a six-month measurement before
253 dosing were included. To prevent differences in cohort, site, intervention or randomized-group composition
254 from driving the gradient, weight quarters were assigned within cohort $\times$ site $\times$ intervention $\times$ treatment group.
255 Direct-standardization weights then gave every eligible arm $\times$ site $\times$ treatment cell the same relative
256 contribution in each quarter. Weighted Kaplan–Meier medians were calculated for treated and control
257 animals, and proportional gain was $100 \times (\text{treated median} - \text{control median}) / \text{control median}$. A left-truncated
258 Cox model stratified by arm $\times$ site, clustered by mouse and allowing arm-specific treatment effects estimated
259 the shared treatment-by-ordinal-quarter trend. Conventional site-pooled quartiles and heavy-stratum tertile,
260 quartile and fifth analyses were retained as sensitivities.

261 For the comparator-state analysis, each cohort $\times$ site $\times$ sex $\times$ intervention cell was compared with its own
262 contemporaneous control group. Cells required at least 25 treated and 25 control animals, survival beyond the
263 six-month landmark and at least 30 deaths across the two groups. Within each eligible cell, a Cox
264 proportional-hazards model, left-truncated at six months, estimated the treatment log hazard ratio. Each
265 animal appeared once within a cell-level model; control animals could, however, recur across cells when one
266 control group was shared by several intervention arms. Cell-level log hazard ratios were related to the mean
267 six-month weight of the corresponding control group by inverse-variance-weighted regression. The primary
268 within-cohort analysis included cohort fixed effects and standardized control-group weight separately within
269 sex. To assess dependence induced by reused controls, a separate pooled sensitivity analysis used cluster-
270 robust variance at the cohort $\times$ site control-group level (39 clusters). Kaplan–Meier medians were estimated
271 within each cell after left truncation at six months. Percentage median-lifespan gain was calculated within
272 each cell as $100 \times (\text{treated median} - \text{control median}) / \text{control median}$ and summarized as the unweighted
273 mean across cells in each sex-specific comparator-weight tertile. Comparator-weight tertiles were used for
274 descriptive presentation only; inferential analyses treated control-group weight continuously.

275 For the single-compound analysis, the 27-arm set was collapsed by active compound after excluding three
276 combination arms. All 13 remaining compounds were retained without selection by lifespan effect, and the
277 same direct-standardization weights were used to estimate overall, lowest-quarter and highest-quarter
278 Kaplan–Meier medians. Across the 24 single-agent arms, the common-interaction model retained arm-
279 specific treatment and weight main effects and a shared treatment-by-weight term, matching the primary
280 specification.

281 1.4 month weight and adult-gain models

282 GeneNetwork trait ITP_50601 supplied body weight at approximately 42 days in three cohorts; the depositing
283 HET3_ITP record documents it as circa 42 days (puberty; 1.4 months) on NIH-31 diet (LabDiet 5LG6). Early-
284 life and six-month weights were z-standardized within cohort $\times$ site and entered separately and jointly in Cox
285 models left-truncated at six months. The joint male-control model included 539 mice with both measurements.
286 The trajectory analysis included all sixteen cohorts with six- and twelve-month weights among mice surviving
287 beyond day 365. Percentage gain was centred within cohort $\times$ site $\times$ arm $\times$ dose $\times$ sex cells and scaled by the
288 pooled within-cell s.d. separately by sex; cells required at least twelve deaths. Models stratified baseline
289 hazards by cell, used Efron ties and cell-clustered robust variance, and estimated gain before and after
290 adjustment for similarly centred six-month weight. The pre-treatment sensitivity retained controls and
291 interventions beginning at or after twelve months. Maternal-age and parity models used the subset with both
292 covariates.

293 Sex-specific weight–mortality models

294 Among untreated controls, separate sex-specific Cox models related weight at 6, 12, 18 and 24 months to
295 subsequent mortality. Weight was standardized within cohort $\times$ site $\times$ sex and baseline hazards were stratified
296 by cohort and site. As a descriptive consistency analysis, site-specific Spearman correlations between six-
297 month weight and age at natural death were calculated within cohort $\times$ intervention $\times$ sex cells containing at
298 least twelve animals, Fisher-z transformed and pooled across sites with $n - 3$ weights.

299 External population and body-composition context

300 The DRiDO female 20% and 40% CR arms were used as external arm-level contrasts rather than pooled with
301 ITP drug arms.¹⁸ UM-HET3 and Diversity Outbred founder architecture was summarized from the original
302 population descriptions.^{3,19,29} UM-HET3 percentage fat came from the published nine-month MRI substudy.²³
303 DRiDO percentage fat was reconstructed as $100 \times \text{fat tissue mass} / (\text{fat tissue mass} + \text{lean tissue mass})$ from
304 the source figure panels, matching the study's available mass components. Founder-strain weights and ages
305 were taken from Mouse Phenome Database Reed1 measures 32603 and 32680; housing and diet were taken
306 from the corresponding project animal documentation.⁴

307 Cross-study intervention analysis

308 The assembled cross-study corpus contained 324 arms from 89 studies reporting both treatment-associated
309 weight change and lifespan change. Six mixed- or pooled-sex arms from six studies contributed no sex-
310 specific comparison, leaving 318 sex-specific intervention arms from 83 studies for analysis; Supplementary
311 Data 3 lists the complete 324-arm corpus. The responder analysis included 134 arms in which weight
312 decreased and lifespan increased, classified as ITP pharmacological, non-ITP pharmacological or dietary
313 restriction. Within each source-by-sex stratum, percentage lifespan gain was regressed through the origin on
314 percentage body-weight reduction, such that the slope represents lifespan gain per 1% weight reduction, with
315 standard errors clustered on study. Sex and source differences were tested by interactions with the weight
316 term in the same clustered model, and the analysis was repeated after requiring at least 5% weight reduction.
317 Weight and lifespan values followed the source studies and therefore included both mean and median lifespan
318 measures and directly reported or figure-estimated weights.³⁰ Values lacking independent source support, and
319 pre-dosing values representing baseline imbalance rather than treatment response, were excluded before
320 analysis. Because these comparisons are drawn between studies rather than within randomized arms, study-
321 level differences in husbandry, diet and survival ascertainment may confound these cross-study associations,
322 which are therefore interpreted descriptively.

323 Body-composition synthesis

324 Treated and control percentage-fat values were taken from each study's reported definition where available
325 and otherwise digitized from figure panels. When only tissue masses were plotted, percentage fat was
326 reconstructed as fat mass divided by body weight or, for DRiDO, as $100 \times \text{fat tissue mass} / (\text{fat tissue mass} +$
327 $\text{lean tissue mass})$, consistent with the available components. Values derived from lean-volume complements,
328 unmatched doses or sexes, and combined-sex lifespan records were excluded. Pearson and Spearman
329 correlations, study-level aggregation, a study-clustered regression and leave-one-study-out analyses related
330 change in fat percentage to lifespan gain. The 36-arm extraction is provided as Supplementary Data 2.

331 Statistical analysis

332 Cox proportional-hazards models used Efron handling of ties. All tests were two-sided. Interaction estimates
333 are reported as HRs per +1 s.d. of weight, except for the ordinal matched-quartile trend, which is per heavier
334 quarter. Weighted or conventional Kaplan–Meier medians were used for direct lifespan contrasts as specified.
335 Primary inference rests on the unselected all-arm interaction; successful-arm, matched-quartile, comparator,
336 developmental, cross-study and body-composition analyses are complementary. Analyses were performed in
337 R and Python using survival, pandas and matplotlib.

338 Data availability

339 Individual ITP body weights and survival are publicly available from the Mouse Phenome Database ITP
340 portal (<https://phenome.jax.org/centers/ITP>; projects ITP1 and ITP2004–ITP2021) and the GeneNetwork

341 HET3-ITP trait resource (https://genenetwork.org/show_trait?dataset=HET3-ITPPublish&trait_id=50601;
342 trait ITP_50601). Supplementary Data 1–6 provide, respectively, the harmonized ITP mouse-level file, body-
343 composition extraction, cross-study intervention table, arm-level contrasts, stratum correlations and matched-
344 quartile results. Founder-strain comparison data are available from the Mouse Phenome Database Reed1
345 project (<https://phenome.jax.org/projects/Reed1>; measures 32603 and 32680).

346 Code availability

347 Executable scripts and input data for the principal analyses are provided in the accompanying reproducibility
348 package, including the primary all-arm and weight-lowering-class models, landmark and matched-quartile
349 analyses, single-compound and comparator analyses, developmental and longitudinal models, cross-study
350 and body-composition analyses, and founder-strain comparisons. The unselected 26,717-mouse all-arm
351 analysis is the primary estimand.

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Author contributions

The author conceived the study, performed the analyses, interpreted the results and wrote the manuscript.

Competing interests

The author declares no competing interests.

Ethics approval

This secondary analysis used publicly available animal-study data; no new animal experiments were performed.

Tables

Table 1 | Treatment-by-weight moderation estimates

Analysis	Sex	Arms	Mice	Interaction HR (95% CI)	P
All arms (primary)	Male	89	14,301	0.95 (0.91–0.99)	0.012
All arms (primary)	Female	85	12,416	1.02 (0.98–1.06)	0.29
Weight-lowering	Male	39	8,022	0.90 (0.85–0.94)	3.6×10^{-5}

Analysis	Sex	Arms	Mice	Interaction HR (95% CI)	P
Non-lowering	Male	50	9,270	0.99 (0.95–1.04)	0.81
Weight-lowering	Female	37	7,061	1.00 (0.95–1.05)	0.95
Non-lowering	Female	48	8,228	1.04 (0.99–1.09)	0.12
Pre-dosing, all arms	Male	56	10,636	0.96 (0.92–1.01)	0.11
Pre-dosing, weight-lowering	Male	27	6,080	0.91 (0.85–0.96)	0.0011
Published extenders at 12 months	Male	21	4,983	0.91 (0.86–0.97)	0.0042
Published extenders at 12 months	Female	11	2,606	0.98 (0.91–1.07)	0.68
1.4-month weight	Male	4	730	1.00 (0.85–1.17)	0.97
1.4-month weight	Female	2	553	1.03 (0.86–1.22)	0.77

HR < 1 indicates that treatment became more favourable at higher weight. In joint mouse-clustered models, the male class contrast was HR 0.90 (95% CI 0.85–0.95; P = 2.9 × 10⁻⁴); in the pre-dosing subset it was HR 0.89 (P = 0.0014). Mice can contribute to more than one arm through shared controls; variance was clustered by mouse.

Table 2 | Median lifespan by quarter of pre-treatment body weight in male weight-lowering arms

Baseline weight	Mean control weight	Control	Treated	Gain
Lightest quarter	34.8 g	892	951	+6.6%
Second quarter	39.7 g	829	913	+10.1%
Third quarter	42.8 g	763	859	+12.6%
Heaviest quarter	47.1 g	673	810	+20.4%

Directly standardized Kaplan–Meier medians in days across 27 male arms (6,080 unique mice). Quartiles were formed within cohort × site × intervention × treatment group from six-month weight measured before dosing; direct-standardization weights equalized the contribution of eligible arm × site × treatment cells across quarters. The treatment-by-ordinal-quarter trend was HR 0.92 (95% CI 0.87–0.97; P = 0.0011). Conventional site-pooled gains were 5.2%, 12.7%, 14.5% and 19.5%. Across 29 non-lowering male arms the corresponding gains were 4.2%, 1.1%, 4.4% and 4.7%.

Table 3 | Median-lifespan gain across pre-treatment weight strata for all single compounds

Compound	Arms	Mice	Lowest-weight quarter	Overall	Highest-weight quarter
16α-hydroxyestradiol	1	452	+17.7%	+15.3%	+25.7%
Acarbose	3	789	+7.3%	+14.3%	+28.4%
17α-estradiol	4	1,427	+5.9%	+14.1%	+22.9%
Canagliflozin	2	836	+6.1%	+13.0%	+23.2%
Rapamycin	6	1,293	+6.7%	+9.5%	+11.6%
Meclizine	1	306	+2.6%	+8.1%	+12.4%
Fisetin	1	311	−7.4%	+6.2%	+6.8%
Glycine	1	291	+4.0%	+5.9%	+3.8%
MIF098	1	433	+6.5%	+5.6%	+14.7%
Candesartan	1	434	+2.2%	+5.0%	+24.1%
Geranylgeranylacetone	1	431	−2.8%	+4.2%	+19.9%
NDGA	1	150	+10.9%	+2.5%	+11.3%
Dimethyl fumarate	1	306	−5.3%	−0.8%	+11.9%
Pooled single-agent set	24	5,523	+5.2%	+10.0%	+20.9%

The 24 single-agent arms in the pre-treatment weight-lowering population were collapsed by active compound; three combination arms were excluded. Every represented compound is shown. Values are directly standardized percentage differences in left-truncated Kaplan–Meier median lifespan. The pooled row standardizes across all eligible arms rather than averaging compound rows.

Table 4 | Size of the treatment effect against the lowest-weight and highest-weight comparator groups

Effect measure	Lightest comparators	Heaviest comparators	Fold change
Males, reduction in hazard	11.0%	22.3%	2.0
Females, reduction in hazard	4.3%	15.1%	3.5
Males, median lifespan gain	4.1%	9.3%	2.3
Females, median lifespan gain	1.6%	4.1%	2.6

Cohort × site × sex × arm cells (426 cells; 16 cohorts) were grouped into tertiles by the mean six-month body weight of their own control group. Reduction in hazard is 1 minus the pooled treatment HR. The male gradient persisted within cohort (P = 0.0081) but was attenuated in the shared-control-clustered sensitivity analysis (P = 0.078); neither female analysis supported a gradient (P = 0.63 and P = 0.37). Median-lifespan gain is the unweighted mean of within-cell percentage gains in six-month-left-truncated Kaplan–Meier medians.

448 Extended Data Tables

449 Extended Data Table 1 | Developmental and age-specific weight–mortality estimates in untreated controls

Exposure/model	Sex	n	Mortality HR (95% CI)	P
6-month attained weight	Male	3,799	1.39 (1.34–1.44)	2.3×10^{-79}
6-month attained weight	Female	3,543	1.08 (1.04–1.11)	1.4×10^{-5}
12-month attained weight	Male	3,469	1.36 (1.31–1.41)	6.5×10^{-59}
12-month attained weight	Female	3,451	1.09 (1.05–1.13)	5.3×10^{-7}
6-to-12-month gain, unadjusted for 6-month weight	Male	11,300	0.97 (0.95–0.99)	0.018
6-to-12-month gain, adjusted for 6-month weight	Male	11,300	1.09 (1.06–1.12)	2.4×10^{-10}
6-to-12-month gain, adjusted for 6-month weight	Female	10,521	1.03 (1.01–1.05)	9.0×10^{-4}
Pre-treatment sensitivity, gain adjusted for 6-month weight	Male	5,393	1.09 (1.05–1.13)	2.4×10^{-6}

450 HRs are per +1 s.d. standardized within the specified analysis cell. Attained-weight rows use untreated controls. Longitudinal weight-gain rows include
 451 all eligible arms; gain was centred within cohort $\times$ site $\times$ arm $\times$ dose $\times$ sex, such that treated animals contributed only through within-arm variation.
 452 Full gain models begin follow-up at twelve months and adjust for six-month attained weight except where stated.

453 Extended Data Table 2 | Maternal-age and parity sensitivity

Sex	n	Weight HR	Weight HR after maternal adjustment
Male	286	1.403 (1.238–1.591)	1.402 (1.233–1.595)
Female	251	1.145 (0.994–1.320)	1.137 (0.985–1.311)

454 Six-month weight association before and after adjustment for maternal age and parity.

455 Extended Data Table 3 | Association between change in body-fat percentage and lifespan gain

Analysis	Arms/studies	Estimate (95% CI)	P
Pearson correlation	36 / 14	$r = -0.10$ (–0.41 to 0.24)	0.58
Spearman correlation	36 / 14	$\rho = 0.08$	0.66
Study-level correlation	14 studies	$r = -0.21$	0.47
Study-clustered slope	36 / 14	-0.17 (–0.74 to 0.40)	0.57

456 Body-fat change is treated minus control in percentage points; lifespan gain is the percentage increase relative to control. Directly reported percentage
 457 fat was preferred over reconstructed values.

458 Extended Data Table 4 | Weight reduction and lifespan gain by study source and sex

Intervention source	Sex	Arms	Studies	Slope (95% CI)	Median ratio
ITP pharmacological	Female	25	12	0.46 (0.17–0.75)	0.40
ITP pharmacological	Male	31	13	1.03 (0.75–1.32)	1.27
Non-ITP pharmacological	Female	22	15	0.93 (0.55–1.31)	1.00
Non-ITP pharmacological	Male	29	23	0.79 (0.40–1.18)	1.29
Dietary restriction	Female	15	7	0.79 (0.62–0.97)	0.67
Dietary restriction	Male	12	8	0.81 (0.68–0.94)	0.74
Sex difference: ITP	Female–male	56	—	-0.57 (–0.98 to –0.16)	$P = 0.0063$
Sex difference: non-ITP pharmacological	Female–male	51	—	0.14 (–0.31 to 0.60)	$P = 0.54$
Sex difference: dietary restriction	Female–male	27	—	-0.02 (–0.26 to 0.23)	$P = 0.90$

459 Slopes are percentage lifespan gain per 1% lower body weight among the 134 responder arms. The ITP female–male difference was -0.57 ($P =$
 460 0.0063), compared with 0.14 ($P = 0.54$) outside ITP; the pharmacological sex $\times$ source interaction was significant ($P = 0.021$) and remained significant
 461 after requiring at least 5% weight loss ($P = 0.018$).

462 Extended Data Table 5 | Population conditions

Feature	ITP / UM-HET3	DRiDO / Diversity Outbred
Founder architecture	Four inbred founders	Eight founders, including three wild-derived
Intervention structure	Pharmacological arms	Caloric restriction
Classical founders	BALB/cByJ, C57BL/6J, C3H/HeJ, DBA/2J	Five classical plus CAST, PWK and WSB
Obesity/diabetes-model founders	None dedicated	NZO/HILtJ and NOD/ShiLtJ

463 Founder architecture and intervention context of the ITP/UM-HET3 and DRiDO/Diversity Outbred populations.

464 Extended Data Table 6 | Male body weight in the four UM-HET3 founder strains

Founder strain	Male weight
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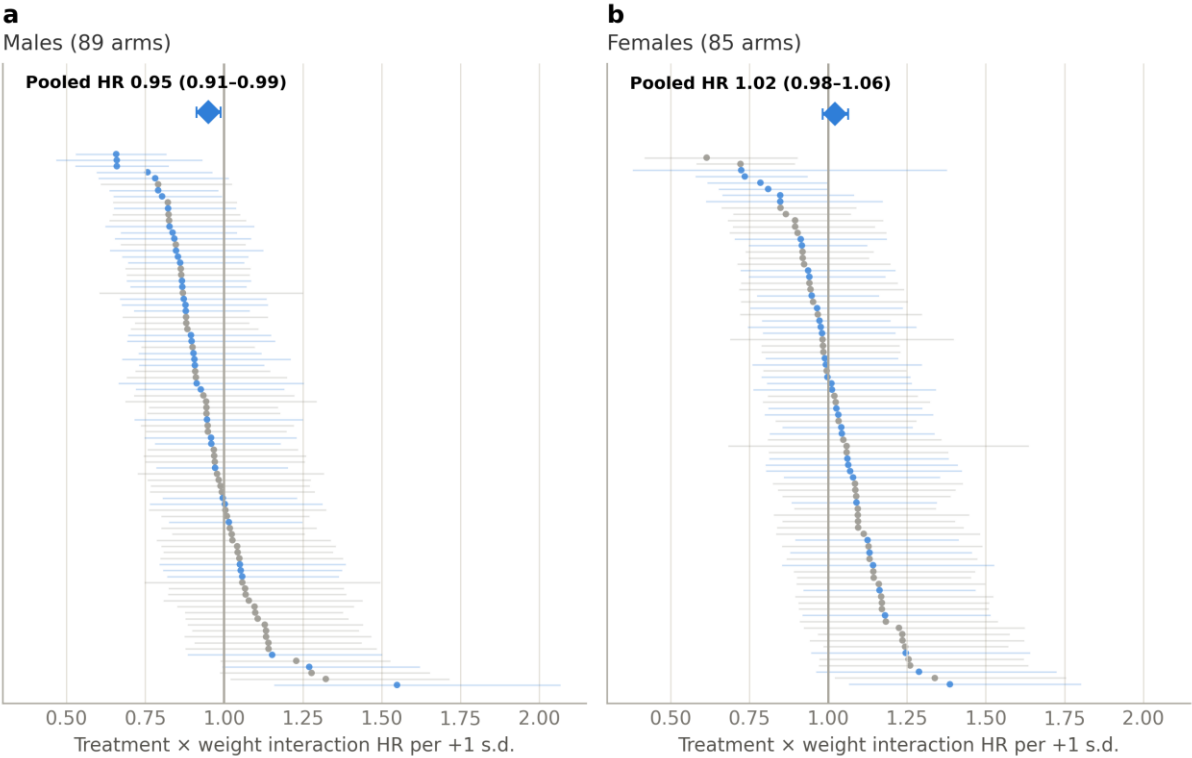
BALB/cByJ	32.0 g
C57BL/6J	30.8 g
C3H/HeJ	26.7 g
DBA/2J	28.1 g
Four-founder mean	29.4 g

Unadjusted male means from Mouse Phenome Database Reed1 measure 32603; corresponding mean ages were approximately 30.3–31.5 weeks from measure 32680 (N = 10–12 per strain).⁴ Mice were individually housed with ad-libitum access to Teklad Rodent Diet 8604. The four-founder mean is the unweighted mean of the strain estimates.

Figure legends and figures

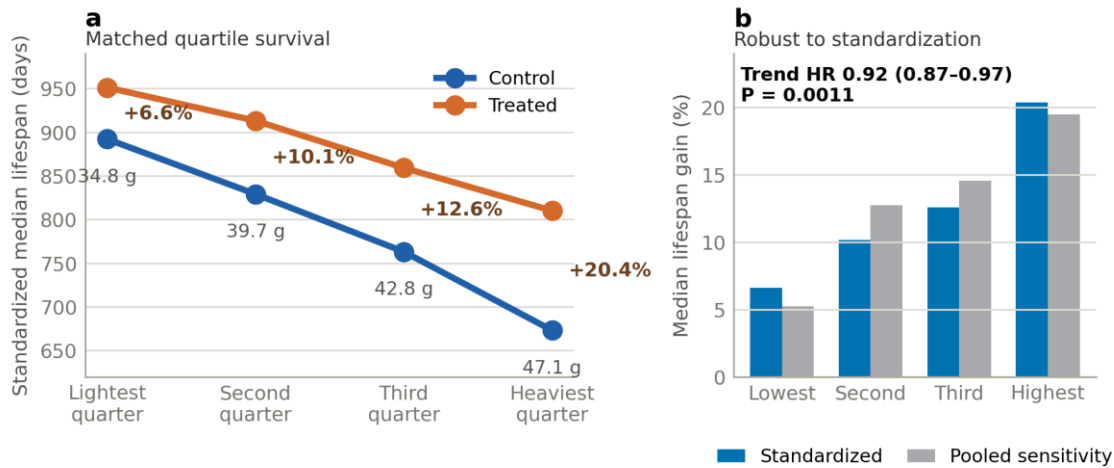
Figure 1 | Higher within-arm adult weight predicted greater treatment benefit in males.

Arm-specific within-arm interaction estimates. a, 89 male arms. b, 85 female arms. Adult weight was standardized within treatment group × site × sex, removing each intervention's average weight effect. Diamonds show pooled sex-specific estimates. Blue marks the 39 male and 37 female weight-lowering arms and grey the remainder; an arm was classified as weight-lowering when treated animals averaged more than 1 g below their own controls after dosing began.



475 **Figure 2 | Pre-treatment adult weight amplified lifespan benefit**

476 a, Arm- and site-standardized Kaplan–Meier median lifespan by matched pre-dosing weight quarter in 27 male weight-lowering arms. b,
477 Standardized gains compared with a conventional site-pooled sensitivity analysis. Quartiles were formed within cohort × site × intervention
478 × treatment group; the ordinal treatment-by-quarter trend was HR 0.92 (95% CI 0.87–0.97; P = 0.0011).

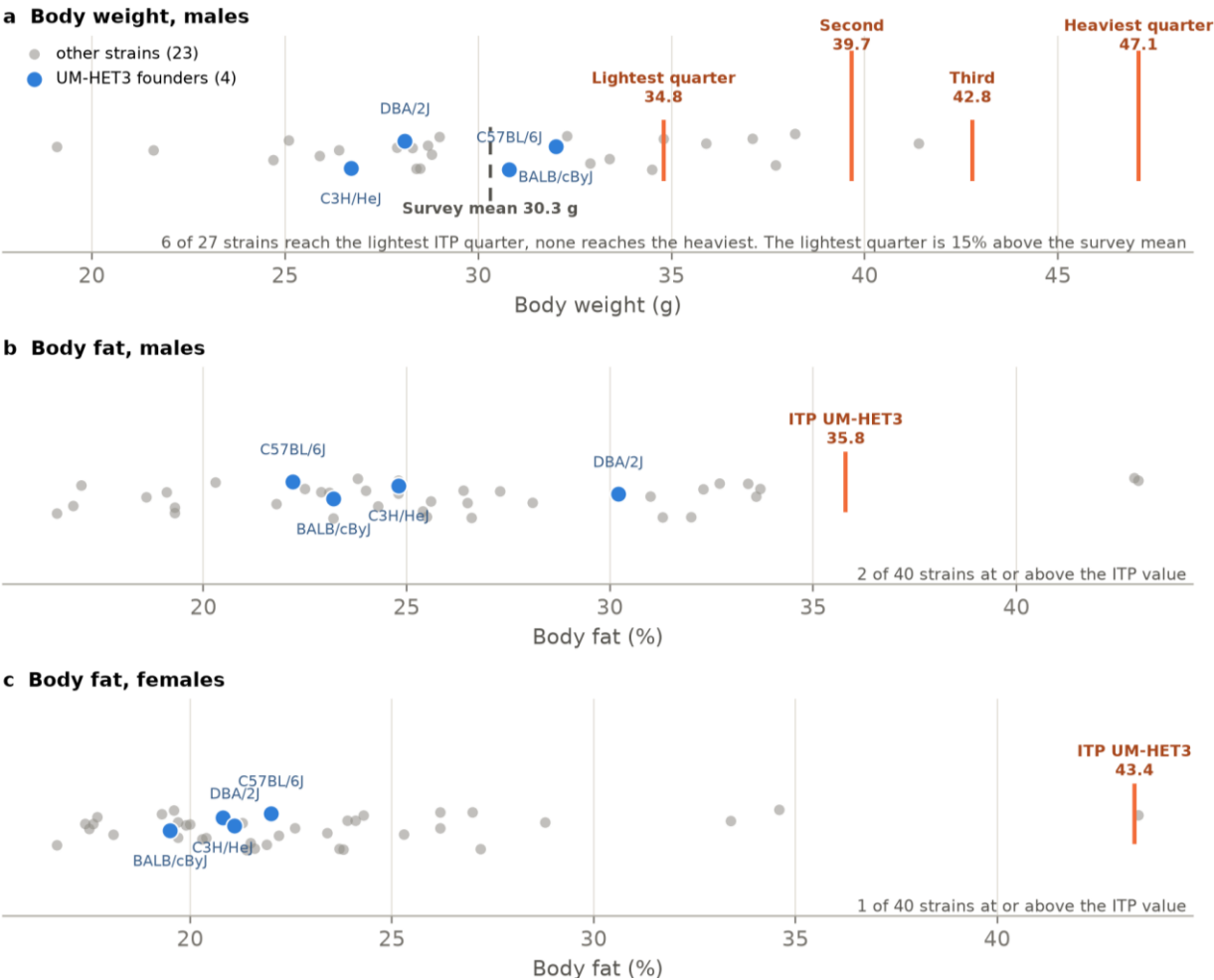


479

480 **Figure 3 | Adult body weight and adiposity in ITP and laboratory-strain references**

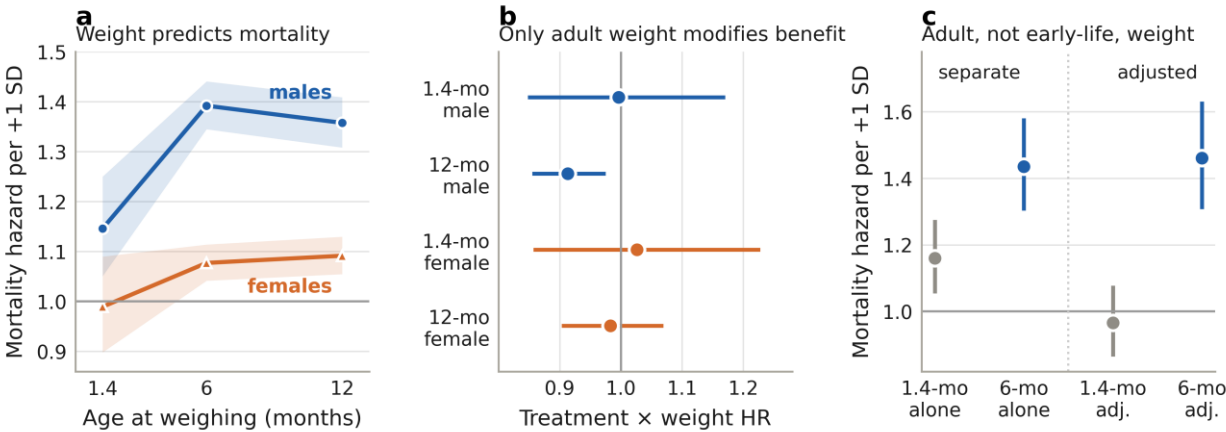
481 Each grey point represents one inbred-strain mean; the four UM-HET3 founder strains are shown in blue and ITP values on the same axis. **a**,
482 Male body weight, with the four matched pre-treatment ITP weight quarters and laboratory-strain mean indicated. **b**, Male body fat. **c**, Female
483 body fat. Mouse Phenome Database values are from the Reed1 body-weight survey ⁴ (27 of 28 strains measured at 29–34 weeks) and the 40-
484 strain dual-energy X-ray absorptiometry survey ²² (14–18 weeks). ITP weights are control means at six months, and ITP body fat was

485 measured by whole-body MRI at nine months ²³; ages are therefore not matched between panels. Founder-strain weights are reported in
486 Extended Data Table 6.



487
488 **Figure 4 | The signal emerged with attained adult weight**

489 a, Subsequent mortality HRs for weight measured at 1.4, 6 and 12 months in untreated controls. b, Treatment-by-weight moderation at 1.4
490 and 12 months among published lifespan-extending arms. c, 1.4- and 6-month weight in separate and mutually adjusted models among 539
491 male controls with both measurements. Blue denotes males and orange females.



492